Functions act on paths: for f from A to B there is an operation from x = y to f(x) = f(y) carrying refl to refl

The Univalent Foundations Program

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Editorial note. The counted statement is the author lemma labelled lem:map (source lines 693-704); the proof span 708-717 is the author's first proof environment following it, retained line for line.

Editorial note. The author gives a second proof at source lines 719-723; it is not counted and not retained on the card face.

Editorial note. The statement and proof cite no external labels, so no external references are declared.

Editorial scope. One original result of the HoTT book chapter cha:basics, counted once: the action of a function on paths (functoriality of ordinary functions on identity types). Only the counted lemma and its first author proof are retained; the author's second proof and the notation commentary remain uncounted in the source.

Allowed standard prerequisites (editorial).

- The identity type of a type A and its reflexivity element refl x.
- The induction principle for identity types (path induction) with its conversion rule.
- Ordinary (non-dependent) functions f from A to B and the identity type of B.

Lemma 0.0.1. *Suppose that $f : A \to B$ is a function. Then for any $x, y : A$ there is an operation*

$$\mathsf{ap}_f : (x =_A y) \to (f(x) =_B f(y)).$$

Moreover, for each $x : A$ we have $\mathsf{ap}_f(\mathsf{refl}_x) \equiv \mathsf{refl}_{f(x)}$.

First proof. Let $D : \prod_{(x,y:A)}(x = y) \to \mathcal{U}$ be the type family defined by

$$D(x, y, p) :\equiv (f(x) = f(y)).$$

Then we have

$$d :\equiv \lambda x.\, \mathsf{refl}_{f(x)} : \prod_{x:A} D(x, x, \mathsf{refl}_x).$$

By path induction, we obtain $\mathsf{ap}_f : \prod_{(x,y:A)}(x = y) \to (f(x) = f(y))$. The computation rule implies $\mathsf{ap}_f(\mathsf{refl}_x) \equiv \mathsf{refl}_{f(x)}$ for each $x : A$. □